

# Airline Satisfaction Analysis

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2024-12-23

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## 1. Introduction

To complement this report, an ANNEX document is included, containing all the results, visualizations, and plots necessary to interpret and validate the findings. This additional resource provides a thorough understanding of the analysis and facilitates the reproducibility of the study.

The primary objective of this project is to predict customer satisfaction probabilities using the training and test samples from the Airline Passengers Satisfaction dataset available on Kaggle. Additionally, we aim to interpret the final binary outcome model to identify which variables most significantly influence customer satisfaction decisions.

The project begins with a data preparation and exploratory analysis phase, which focuses on ensuring data quality. This includes generating a detailed Data Quality Report, identifying and imputing missing values,

and examining variable distributions and relationships. A univariate descriptive analysis is conducted to assess the characteristics of individual variables.

In the modeling phase, we utilize the training dataset to build and refine predictive models, followed by an evaluation of model quality and predictive performance on the test dataset. The steps in this process include:

- Profiling and Feature Selection: Identifying key variables that contribute to the predictive power of the model.
- Residual Analysis: Detecting and filtering unusual and influential data points.
- Numerical Variable Transformations: Applying transformations to improve model fit and address data skewness.
- Factor Effects and Interactions: Adding main effects and second-order interactions involving categorical variables to enhance the model.
- Iterative Residual Analysis: Reassessing residuals and influential data iteratively to refine the model.
- Goodness of Fit and Model Interpretation: Evaluating model performance and interpretability using both the training and test datasets.

## 2. Data Preparation

### 2.1 Data loading

### 2.2 Univariate Descriptive Analysis

In this section, we undertake a crucial step in any data science project, particularly for modeling purposes. A detailed descriptive analysis is performed on each variable, highlighting their most relevant characteristics. We follow a column-by-column examination of the data set, and then generate a data quality report and prepare the data for the subsequent modeling stage by addressing missing values through imputation methods, and deleting non-informative variables and creating new ones.

Note that the explanatory data analysis is assessed using exclusively the train data, even though new variables and variable removals also impact the test data set. In this initial phase, we analyze all variables, highlighting their most significant characteristics. We also convert numeric variables that represent qualitative concepts into factors and consider creating new variables based on existing numeric ones. Not only that but also ordinal numerical variables like ratings will be converted into ordered factors. Subsequently, new induced factor variables like age-brackets will also be introduced in this section.

#### 2.2.1 Variable Gender

*Gender* is a binary categorical variable representing the passenger's gender. The distribution is nearly balanced, with a slightly higher proportion of females compared to males, and there are no missing values.

```
summarize_categorical_data(train$Gender)
summarize_categorical_data(test$Gender) # 0 NAs
```

#### 2.2.2 Variable Customer.Type

**Customer.Type** is a binary categorical variable indicating customer loyalty status (Loyal or Disloyal). The distribution is highly imbalanced, with 81% classified as Loyal and 19% as Disloyal. No missing values are present.

```
summarize_categorical_data(train$Customer.Type)
summarize_categorical_data(test$Customer.Type) # 0 NAs
```

### 2.2.3 Variable Age

**Age** is a numerical variable representing the age of customers, ranging from 7 to 85 years, with a mean of 39.41 and a standard deviation of 14.92. Despite the large sample size, both visual inspections (histogram and boxplot) and statistical tests strongly indicate no normality. No missing values or outliers are present in the variable.

```
summary_and_normality_check(train$Age,bins=25)
summary_and_normality_check(test$Age,bins=25) # 0 NAs
```

### 2.2.4 Variable Type.of.Travel

**Type.of.Travel** is a categorical binary variable that represents the kind of flight of the observation (Business or Personal). The proportions are skewed towards business travel (0.692 vs 0.308) and there are no null values.

```
summarize_categorical_data(train$Type.of.Travel)
summarize_categorical_data(test$Type.of.Travel) # 0 NAs
```

### 2.2.5 Variable Class

**Class** categorical ternary variable that represents the type of cabin travel class of the observations (Business, Eco or Eco Plus). Vast majority of observations belong to Business or Eco evenly, and Eco plus has belongs to 7.22% of the observations. There are no null values.

```
summarize_categorical_data(train$Class)
summarize_categorical_data(test$Class) # 0 NAs
```

### 2.2.6 Variable Flight.Distance

**Flight.Distance** is a discrete finite numerical variable that represents the distance of the given flight in a range between [67, 4983] with a mean value of 1194 and 999 std. Its distribution is obviously not normal given the almost exponential shape that it has and both inferential tests reject with tiny p-values. There are 95 mild outliers, none of them severe, and there is no metadata about the units. There are no null values.

```
outliers_indexes <- summary_and_normality_check(train$Flight.Distance, bins=10)
outliers_indexes
all_outliers[outliers_indexes] <- all_outliers[outliers_indexes] + 1
summary_and_normality_check(test$Flight.Distance, bins=10) # 0 NAs
```

### 2.2.7 Variable Inflight.wifi.service

**Inflight.wifi.service** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The majority of values are grouped between 2 and 3, with a mean of 2.7, median of 3 and deviation of 1.26, producing a uni modal bell-shaped distribution centered at 2-3. There are 149 null values in the training set.

```
train$Inflight.wifi.service <- summarize_ordinal_data(train$Inflight.wifi.service)
test$Inflight.wifi.service <- summarize_ordinal_data(test$Inflight.wifi.service) # 58 NAs
```

### 2.2.8 Variable Departure.Arrival.time.convenient

**Departure.Arrival.time.convenient** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The majority of values are grouped between 4 and 5, with a mean of 3.22, median of 3 and deviation of 1.38. The distribution is right-skewed and has a heavy left tail. There are 247 values equal to 0, we consider them as Not Applicable (NA).

```
train$Departure.Arrival.time.convenient <-
  summarize_ordinal_data(train$Departure.Arrival.time.convenient)
test$Departure.Arrival.time.convenient <-
  summarize_ordinal_data(test$Departure.Arrival.time.convenient) # 115 NAs
```

### 2.2.9 Variable Ease.of.Online.booking

**Ease.of.Online.booking** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The distribution of ratings is somewhat bell-shaped and unimodal, with a mean of 2.89, median 3 and deviation of 1.29. There are 127 null values.

```
train$Ease.of.Online.booking <-
  summarize_ordinal_data(train$Ease.of.Online.booking)
test$Ease.of.Online.booking <-
  summarize_ordinal_data(test$Ease.of.Online.booking) # 85 NAs
```

### 2.2.10 Variable Gate.location

**Gate.location** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The distribution is somewhat uniform with a median of 3, mean of 3 and deviation of 1.28. There are no null values.

```
train$Gate.location <- summarize_ordinal_data(train$Gate.location)
test$Gate.location <- summarize_ordinal_data(test$Gate.location) # 0 NAs
```

### 2.2.11 Variable Food.and.drink

**Food.and.drink** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The distribution is uniform and right skewed, with a median of 3, mean of 3.19 and deviation of 1.33. There are 5 null values.

```
train$Food.and.drink <- summarize_ordinal_data(train$Food.and.drink)
test$Food.and.drink <- summarize_ordinal_data(test$Food.and.drink) # 1 NAs
```

### 2.2.12 Variable Online.boarding

**Online.boarding** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The barplot showcases a fairly unimodal right skewed distribution, with a median of 4, mean of 3.32 and deviation of 1.26. There are 119 null values.

```
train$Online.boarding <- summarize_ordinal_data(train$Online.boarding)
test$Online.boarding <- summarize_ordinal_data(test$Online.boarding) # 47 NAs
```

### 2.2.13 Variable Seat.comfort

**Seat.comfort** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is right skewed with a heavy left tail, and has a median of 4, mean of 3.41 and a deviation of 1.33. There are no null values.

```
train$Seat.comfort <- summarize_ordinal_data(train$Seat.comfort)
test$Seat.comfort <- summarize_ordinal_data(test$Seat.comfort) # 0 NAs
```

### 2.2.14 Variable Inflight.entertainment

**Inflight.entertainment** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is unimodal and slightly right-skewed, with a median of 4, mean 3.34 and deviation of 1.33. There is 1 null value.

```
train$Inflight.entertainment <-
  summarize_ordinal_data(train$Inflight.entertainment)
test$Inflight.entertainment <-
  summarize_ordinal_data(test$Inflight.entertainment) # 0 NAs
```

### 2.2.15 Variable On.board.service

**On.board.service** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Similarly, its distribution is unimodal and right skewed with a median of 4, mean of 3.37 and deviation of 1.27. There is one null value.

```
train$On.board.service <- summarize_ordinal_data(train$On.board.service)
test$On.board.service <- summarize_ordinal_data(test$On.board.service) # 0 NAs
```

### 2.2.16 Variable Leg.room.service

**Leg.room.service** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is more uniform than previous variables but its also somewhat right skewed, with a median of 4 and mean of 3.35, deviation 1.30. There are 22 null values.

```
train$Leg.room.service <- summarize_ordinal_data(train$Leg.room.service)
test$Leg.room.service <- summarize_ordinal_data(test$Leg.room.service) # 8 NAs
```

### 2.2.17 Variable Baggage.handling

**Baggage.handling** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is unimodal and highly right-skewed, having a median of 4, mean of 3.62 and small deviation of 1.18. There are no null values.

```
train$Baggage.handling <- summarize_ordinal_data(train$Baggage.handling)
test$Baggage.handling <- summarize_ordinal_data(test$Baggage.handling) # 0 NAs
```

### 2.2.18 Variable Checkin.service

**Checkin.service** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. In contrast, this distribution seems more uniform with a huge mode between 3 and 4 (median of 3, mean of 3.27, deviation 1.26). There are no null values.

```
train$Checkin.service <- summarize_ordinal_data(train$Checkin.service)
train$Checkin.service <- summarize_ordinal_data(train$Checkin.service) # 0 NAs
```

### 2.2.19 Variable Inflight.service

**Inflight.service** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is unimodal and highly right-skewed, with a median of 4, mean of 3.61 and a small deviation 1.16. There is 1 null value.

```
train$Inflight.service <- summarize_ordinal_data(train$Inflight.service)
test$Inflight.service <- summarize_ordinal_data(test$Inflight.service) # 0 NAs
```

### 2.2.20 Variable Cleanliness

**Cleanliness** is a self-describing ordinal categorical variable given by the customers in the range [1,5]. This variable has the most uniform distribution of them all, median 3, mean 3.26 and deviation 1.3, thus skewed slightly to the right. Again, there are no null values.

```
train$Cleanliness <- summarize_ordinal_data(train$Cleanliness)
test$Cleanliness <- summarize_ordinal_data(test$Cleanliness) # 0 NAs
```

### 2.2.21 Variable Departure.Delay.in.Minutes

**Departure.Delay.in.Minutes** is a discrete numerical finite variable that represents how many minutes this rated flight has been delayed, considering its scheduled departure. Its values range from 0 to 692, with a mean of 15, median of 0 and huge deviation of 38.45. There are 675 mild outliers and 413 severe ones. Its distribution looks astonishingly exponential, with a skewness coefficient of 5.99. Both inferential tests reject the hypothesis of normality.

```
outliers_indexes_dep_delay <-
  summary_and_normality_check(train$Departure.Delay.in.Minutes)
all_outliers[outliers_indexes] <- all_outliers[outliers_indexes_dep_delay] + 1
table(all_outliers)
```

### 2.2.22 Variable Arrival.Delay.in.Minutes

**Arrival.Delay.in.minutes** is a discrete numerical finite variable that represents how many minutes this rated flight has been delayed, considering its scheduled arrival. Similarly to departure delay, its distribution has an astonishingly exponential shape, and a huge coefficient of skewness of 6.06. Has a mean of 15.28 but a median of 0, with values ranging between [0, 702] minutes, 12 null values, 669 mild outliers and 410 severe ones. Both inferential tests reject the hypothesis of normality.

```

outliers_indexes_arr_delay <-
  summary_and_normality_check(train$Arrival.Delay.in.Minutes)
all_outliers[outliers_indexes] <- all_outliers[outliers_indexes_arr_delay] + 1
table(all_outliers)

```

### 2.2.23 Variable satisfaction

**satisfaction** is a target binary categorical variable that represents whether the customer is satisfied or not/neutral with its overall experience. The dataset is somewhat balanced with a slight bias towards neutral/dissatisfied (57.14% vs 42.86%). There are no null values.

```

summarize_categorical_data(train$satisfaction)
summarize_categorical_data(test$satisfaction)

```

### Transform variables

Here, we transform variable types to their correct ones.

```

str(train)
# factor variables
train[, c(1,2,4,5,23)] <- lapply(train[, c(1,2,4,5,23)], as.factor)
test[, c(1,2,4,5,23)] <- lapply(test[, c(1,2,4,5,23)], as.factor)

# numerical variables
train[, c(3,6,21,22)] <- lapply(train[, c(3,6,21,22)], as.numeric)
test[, c(3,6,21,22)] <- lapply(test[, c(3,6,21,22)], as.numeric)

# ordinal variables
train[, c(7:20)] <- lapply(train[, c(7:20)],
  function(x) factor(x, levels = c(1, 2, 3, 4, 5),
    ordered = TRUE))
test[, c(7:20)] <- lapply(test[, c(7:20)],
  function(x) factor(x, levels = c(1, 2, 3, 4, 5),
    ordered = TRUE))

str(train)

```

### 2.2.23 New variables definition

We defined 4 new variables that we think that might be improving our model since can provide information on the relation between variables and the satisfaction of an individual. These consist of:

- AgeGroup: Divide ages in 4 different groups
- f.FlightDist: Divide distances of flights in 3 different groups
- DelayedArrival: If the arrival time is delayed
- DelayedDeparture: If the departure is delayed.

```

# age factor
train$AgeGroup <- cut(train$Age, breaks = c(-Inf, 18, 35, 60, Inf),
  labels = c("Under 18", "18-35", "36-60", "Over 60"))

```

```

test$AgeGroup <- cut(test$Age, breaks = c(-Inf, 18, 35, 60, Inf),
                     labels = c("Under 18", "18-35", "36-60", "Over 60"))

# distance factor
quant <- quantile(train$Flight.Distance, probs = seq(0, 1, 0.333), na.rm = TRUE)
train$f.FlightDist <- cut(train$Flight.Distance,
                          breaks = quant,
                          labels = c("Short Distance", "Medium Distance",
                                     "Long Distance"),
                          include.lowest = TRUE)

quant2 <- quantile(test$Flight.Distance, probs = seq(0, 1, 0.333), na.rm = TRUE)
test$f.FlightDist <- cut(test$Flight.Distance,
                         breaks = quant2,
                         labels = c("Short Distance", "Medium Distance",
                                    "Long Distance"),
                         include.lowest = TRUE)

# delay arrival (binary)
train$DelayedArrival <- as.factor(ifelse(train$Arrival.Delay.in.Minutes > 0,
                                         "DelayedArrival", "OK"))
test$DelayedArrival <- as.factor(ifelse(test$Arrival.Delay.in.Minutes > 0,
                                         "DelayedArrival", "OK"))
summary(train$DelayedArrival)

train$DelayedDeparture <- as.factor(ifelse(train$Departure.Delay.in.Minutes > 0,
                                         "DelayedDeparture", "OK"))
test$DelayedDeparture <- as.factor(ifelse(test$Departure.Delay.in.Minutes > 0,
                                         "DelayedDeparture", "OK"))
summary(train$DelayedDeparture)

```

## 2.3 Data quality report

This section provides a comprehensive overview of the dataset's quality, complementing the prior identification of univariate outliers and missing values with a visual summary to highlight key patterns and anomalies. The data quality report is structured into two parts: an analysis regarding the variables and a review of individual records (stating the number of outliers per individual and identifying multivariate outliers).

### 2.3.1 Variables

This first part of the data quality report assesses the quality of the data regarding the variables.

**2.3.1.1 Missing values and Errors** We used the `md.pattern` function from the `mice` library to analyze the distribution of missing values across the dataset. This function provides a visual representation of missing data patterns, allowing us to identify the extent and location of missingness.

Note that we considered errors (and thus set to NA) observations with value 0 in an ordinal variable with range 1-5.

Based on the results, we can conclude that the dataset is relatively clean, with 4590 observations (91.8%) containing complete data. However, certain variables exhibit notable levels of missing values, particularly those on the right side of the plot. `Departure.Arrival.time.convenient` and `Ease.of.Online.Booking` have the highest number of missing values, with 247 and 217, respectively. `Inflight.wifi.service` has 149 missing values, while `Online.boarding` has 119.

In addition, eight variables have fewer than 25 missing values, and 14 variables are completely free of missing data.

To ensure the integrity of the dataset and support robust and unbiased modeling, all missing and erroneous values will be carefully imputed in a way that preserves the dataset's original characteristics and patterns.

```
summary(train)
md.pattern(train, rotate.names = T)
```

**2.3.1.2 Outliers** In this section, we analyze the presence of outliers in the dataset, focusing on the four numerical variables: `Age`, `Flight.Distance`, `Departure.Delay.in.Minutes`, and `Arrival.Delay.in.Minutes`.

`Age`: No outliers were detected in the `Age` variable, indicating that all recorded values fall within a reasonable and expected range for this dataset.

`Flight.Distance`: A total of 95 mild outliers were identified in `Flight.Distance`, but no severe outliers were detected. This indicates that some flights are moderately longer than the majority, which is expected given variations in airline routes. These values are not concerning and reflect natural variability.

`Departure.Delay.in.Minutes`: Shows 675 mild outliers and 413 severe outliers, indicating that a significant number of flights experienced delays longer than the typical range.

`Arrival.Delay.in.Minutes`: Similarly, `Arrival.Delay.in.Minutes` has 669 mild outliers and 410 severe outliers, closely mirroring the outliers in `Departure.Delay.in.Minutes`. This similarity is expected, as the arrival delay is inherently related to the departure delay. A delayed departure often leads to a delayed arrival unless measures such as flight rescheduling or speed adjustments are taken.

## 2.3.2 Individuals

This second part of the data quality report assesses the number of outliers and missing values per individual and identifies which individuals are multivariate outliers.

**2.3.2.1 Missing values and Errors** Out of the total observations in the dataset, 4590 have no missing values across any variable. Additionally, 178 observations contain a missing value in one variable, 113 observations have missing values in two variables, 90 observations have missing values in three variables, and 29 individuals have missing values in 4 variables

```
na_counts <- rowSums(is.na(train))
table(na_counts)
```

**2.3.2.2 Outliers** Since there are only four numerical variables in the dataset, and the `Age` variable does not have any outliers, the maximum number of outliers an observation can have is three. Based on the analysis, the distribution of outliers across observations is as follows: - 4140 observations have no outliers at all. - 297 observations have one outlier. - 547 observations have two outliers, and notably, 545 of these are outliers in both `Departure.Delay.in.Minutes` and `Arrival.Delay.in.Minutes`. This overlap is expected, as these variables are closely related and delays in departure often propagate to arrival delays. - 13 observations have outliers in all three variables.

This analysis highlights that most observations are free of outliers, while the relationship between `Departure.Delay.in.Minutes` and `Arrival.Delay.in.Minutes` contributes significantly to the overlapping outliers.

```
table(all_outliers)
common_indexes <- intersect(outliers_indexes_arr_delay,
                             outliers_indexes_dep_delay)
length(common_indexes) # 558
```

**2.3.2.3 Multivariate Outliers** The *find\_valid\_moutlier* function identifies the largest combination of numeric variables without missing values that successfully executes the Moutlier function. By iterating through combinations of variables, starting with the largest possible subsets, this function ensures that the process for detecting multivariate outliers incorporates the maximum amount of information available.

Since the Moutlier function depends on some stochastic processes, one can get different results when executing it several times. Thus, we comment the results obtained in one single execution, which has been selected carefully to represent the average values.

There are 213 multivariate outliers, which we proceed to characterize in the next part.

```
result <- find_valid_moutlier(train, quantile = 0.95)
res.out <- result$res_mout;
mult.outliers <- result$multoutliers;

#Print number of outliers
length(mult.outliers)
# Plot Multivariate Outliers
dev.off()
plot(res.out$md, res.out$rd)
abline(h=res.out$cuttoff, col="red", lty=2)
abline(v=res.out$cuttoff, col="red", lty=2)
```

Finally, we present the multivariate outlier profiling:

The chi-square test results suggest significant relationships between the multivariate outliers and several categorical variables, such as *Class*, *Type of Travel*, *Satisfaction*, *Customer Type*, *Flight Distance*, and *Seat Comfort*. Multivariate outliers are primarily associated with specific travel class types, such as “Business” and “Eco Plus,” as well as travel categories like “Business Travel” and “Personal Travel.” Additionally, multivariate outliers tend to have distinct satisfaction levels, with the “satisfied” and “neutral or dissatisfied” categories being most common. There are also strong links to online boarding status and comfort ratings, which help separate outlier from non-outlier categories.

From the analysis of quantitative variables, we see strong associations with Flight Distance and Age. Outliers are more likely to be associated with significantly higher or lower flight distances and age groups, with long-distance flights and older travelers being characteristic of the outlier group. Eta-squared values indicate that Flight Distance has the highest effect size (0.248), which suggests it is a key determinant in distinguishing multivariate outliers. Age, with a smaller effect size (0.0057), also plays a role, though its influence is less pronounced.

Profiling the outliers by cluster, we observe distinct patterns. For the “Not M.O.” (Non-Outlier) group, the mean values for Age and Flight Distance are relatively close to the overall means. In contrast, the “Yes M.O.” (Outlier) group shows significant deviations in both variables, with outliers being characterized by extremely high or low Flight Distance values and older Age profiles.

```
bi_mvout_df <- train
#Binary factor if its a multivariate outlier
```

```

bi_mvout_df$mout <- 0
bi_mvout_df[mult.outliers, "mout"] <- 1
bi_mvout_df$mout <- factor(bi_mvout_df$mout, labels=c("Not M.O", "Yes M.O"))
table(bi_mvout_df$mout)

summary(bi_mvout_df[, c("mout")]);

catdes(bi_mvout_df, num.var=28, proba=0.05)

```

At the moment, we do not remove any observation. However, this are potential influential data for the model that we build as follows, so we will assess carefully which data biases too much the model and see if it is wise to remove it from the data set after computing each model version.

## 2.4 Imputation

In this section, all the missing values previously identified are addressed through an imputation process using mice function.

Note that values are imputed both in the training and test sets, as stated in the statement of the project.

```

summary(train)
imputed_data <- mice(train)
df_complete <- complete(imputed_data)
summary(df_complete)

```

Here, we check the correctness of imputed values.

```

library(dplyr)
# Identify columns with NAs
cols_with_na <- names(train)[sapply(train, function(x) any(is.na(x)))]
numeric_cols_with_na <- cols_with_na[sapply(train[cols_with_na], is.numeric)]

# Numeric variables
quantiles_original <- quantile(train$Arrival.Delay.in.Minutes,
                               probs = seq(0, 1, 0.1), na.rm = TRUE)

quantiles_original
quantiles_imputed <- quantile(df_complete$Arrival.Delay.in.Minutes,
                              probs = seq(0, 1, 0.1), na.rm = TRUE)

quantiles_imputed

Boxplot(train$Arrival.Delay.in.Minutes)
Boxplot(df_complete$Arrival.Delay.in.Minutes)

#Categorical variables
categorical <- cols_with_na[!sapply(train[cols_with_na], is.numeric)]

# Loop through each variable
for (var in categorical) {
  # Print the table comparison for both datasets
  print(list(Train = table(train[[var]]), Complete = table(df_complete[[var]])))

  # Create and print the superimposed bar plot

```

```

print(
  ggplot(rbind(data.frame(ServiceLevel = train[[var]], Dataset = "Train"),
               data.frame(ServiceLevel = df_complete[[var]],
                           Dataset = "Complete")),
         aes(x = factor(ServiceLevel), fill = Dataset)) +
  geom_bar(position = "identity", alpha = 0.5) +
  labs(title = paste("Comparison of", var, "Service Level"))
)
}

```

Now that we have checked the validity of the imputed values, we assign them to the main data frame and impute values in the test set.

```

#impute values to the original df
train <- df_complete

#impute test data
imputed_data_test <- mice(test)
df_complete_test <- complete(imputed_data_test)
test <- df_complete_test

```

## 2.5 Profiling analysis and correlations

The target binary variable **satisfaction** splits the data unevenly: 57.14% neutral or dissatisfied vs 42.86% satisfied. Moreover this variable contains no null values at all. To delve deeper into the main differences between these 2 groups, a clustering profiling will be performed. Just as a remainder, the dataset contains 18 categorical variables (14 ordinal, 3 binary, 1 ternary) and 4 numerical, besides the target variable.

Firstly, the **test.chi2** output points out the fact that almost all categorical variables are statistically significantly related to satisfaction, except for **Gender**. The statistical significance of the performed

$$\chi^2$$

tests is huge for all variables except for **Departure.Arrival.time.convenient**, which has the lowest p-value of them all,

$$1.53 \cdot 10^{-2}$$

.

Secondly, the **quanti.var** output points out that all the numerical variables of the dataset (Flight.Distance, Age, Arrival.Delay.in.Minutes and Departure.Delay.in.Minutes) are statistically significant after performing Anova/T-test, being Flight.Distance and Age the most significant variables.

Finally, taking a glance over **quanti** output we can gather numerical insights on the main difference between the 2 groups after performing numerous v-tests: - Those belonging to the “satisfied” group present a slightly above age, below average delays in departure and arrival and a significantly bigger Flight.Distance, compared to the overall averages. - Those belonging to the “neutral or dissatisfied” in contrast have a below average age of 37 and slightly above average delays in both departure and arrival but significantly below average Flight.Distance, compared to the overall averages.

```

res_cat <- catdes(train, proba=0.05, num.var=23)

#Find categorical variables that are significantly related to satisfaction
#Chi^2 test

```

```
res_cat$test.chi2

#Find numerical variables that are significantly related to satisfaction
# Anova/T test
res_cat$quanti.var

#Compare the 2 clusters assessing numerical differences ---> v test
res_cat$quanti
```

Now, Looking at the relationship between numerical variables, an obvious perfect positive correlation is divided: Departure delay and Arrival delay. Other variables have negligible relations and variables Age and Flight.distance share a small positive correlation.

```
corrplot(cor(train[, num_vars]))
```

### 3. Modeling

First, note that we treat the model Nominally, as the response variable *Satisfaction* have no inherent order.

#### 3.1 Modeling with numeric variables & Residual Analysis

To see which numeric variables provide value to the model, we first create the null model as m0.

In the case of the null model, the intercept estimate is  $\hat{\beta}_0 = -0.28757$ . This means that the log-odds of satisfaction when no explanatory variables are included is -0.28757. The corresponding probability  $P(\text{satisfaction} = 1)$  can be calculated by converting the log-odds to a probability:

$$P(\text{satisfaction} = 1) = \frac{1}{1 + e^{-\hat{\beta}_0}} = \frac{1}{1 + e^{0.28757}} \approx 0.428$$

The AIC (Akaike Information Criterion) value is 6831.2.

```
#Null model
m0 <- glm(satisfaction ~ 1, family=binomial, data=train)
summary(m0)
```

Now, we aim to develop the best numerical variable model, balancing both model fit and complexity. It is clear that the *Age* variable contributes meaningfully to the model. Upon inspecting the residuals versus fitted values plot, we observe that the residuals do not exhibit a flat pattern, suggesting that a transformation of the variable might improve the model. Squaring the variable resolves this issue effectively. Although incorporating  $\text{Age}^3$  appears to be a potential improvement based on the AIC analysis, the residuals versus fitted plot reveals a discernible pattern. Therefore, we exclude  $\text{Age}^3$  from the model.

On the other hand *Flight.Distance* also contributes meaningfully to the model. Incorporating the variable to the model it is clearly worth it, since we obtain better AIC and Residual deviance values and the residuals vs fitted plot keeps showing a flat line. Moreover, the Q-Q Residuals plot now shows a distribution that follows the Normal distribution, which is a good indicator of the model. Transforming the variable is not worth, so we keep it plane.

After incorporating these two variables, we evaluate the potential addition of *Departure.Delay.in.Minutes* and *Arrival.Delay.in.Minutes*.

Following the AIC criteria, including *Departure.Delay.in.Minutes* to the existing model it is not worth it. Moreover, the Anova function (which provides insight into the net effect of adding variables to the model), shows that, at a 95% confidence level, including *Departure.Delay.in.Minutes* after *Arrival.Delay.in.Minutes* does not significantly improve the model.

Therefore, the optimal numerical variable model includes the variables *Age*, *Flight.Distance*, and *Arrival.Delay.in.Minutes*. The residuals vs fitted plot shows a flat line, and the Q-Q Residuals plot shows a normal distribution.

The model is :

$$Y = -0.9981 + (22.67)\text{poly}(\text{Age}, 2)_1 - (36.96)\text{poly}(\text{Age}, 2)_2 \\ + (5.877e - 04)\text{Flight.Distance} - (3.163e - 03)\text{Arrival.Delay.in.Minutes}$$

The model's fit statistics are as follows:

- **AIC:** 6064.3
- **Residual deviance:** 6054.3 (on 4995 degrees of freedom)

Moreover, the Variance Inflation Factor (VIF) indicates no problematic multicollinearity among the predictor variables in the model, and the Residuals vs Leverage plot shows no influential data a priori.

```
#Numerical variables
numeric_vars <- names(which(sapply(train, is.numeric)==TRUE));numeric_vars

#####
#Including Age to the model#
#####
m1 <- glm(satisfaction ~ poly(Age,2), family=binomial, data=train)
summary(m1)
Anova(m1)

par( mfrow=c(2,2))
plot( m1, id.n=0 ) # residual vs fitted plot not showing a flat line.

m1.2 <- glm(satisfaction ~ poly(Age,2), family=binomial, data=train)
m1.3 <- glm(satisfaction ~ poly(Age,3), family=binomial, data=train)
anova(m1.2, m1.3, test="Chisq")
plot( m1.2, id.n=0 )
plot( m1.3, id.n=0 ) # residual vs fitted plot not showing a flat line.

m1 <- m1.2
summary(m1)
Anova(m1)

#####
#Including Flight.Distance to the model#
#####
m2 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance, family=binomial,
          data=train)
summary(m2)
Anova(m2)
anova(m1, m2, test="Chisq")
```

```

plot( m2, id.n=0 )

m2.2 <- glm(satisfaction ~ poly(Age,2) + poly(Flight.Distance,2),
            family=binomial, data=train)
summary(m2.2)
Anova(m2.2)
anova(m2, m2.2, test="Chisq")
plot( m2.2, id.n=0 )

#####
#Including Departure.Delay.in.Minutes to the model#
#####

m3 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Departure.Delay.in.Minutes, family=binomial, data=train)
summary(m3)
Anova(m3)
anova(m2, m3, test="Chisq")

#####
#Including Arrival.Delay.in.Minutes to the model#
#####

m4 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Departure.Delay.in.Minutes + Arrival.Delay.in.Minutes,
          family=binomial, data=train)
summary(m4)
Anova(m4)
anova(m2, m4, test="Chisq")
anova(m3, m4, test="Chisq")

#####
#Removing Departure.Delay.in.Minutes of the model --> FINAL MODEL #
#####

m5 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Arrival.Delay.in.Minutes, family=binomial, data=train)
summary(m5)
Anova(m5)
anova(m2, m5, test="Chisq")
anova(m5, m4, test="Chisq")

vif(m5)
par( mfrow=c(2,2))
plot( m5, id.n=0 )

```

### 3.2 Include factor main effects to the best numeric model & Residual Analysis

Now that we have the best model only including numerical variables, we will include factor main effects to try.

```
factors <- names(which(sapply(train, is.numeric)==FALSE));factors
```

```
#####
# Include Gender #
#####
m6 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Arrival.Delay.in.Minutes + Gender, family=binomial, data=train)
summary(m6)
Anova(m6)
anova(m5, m6, test="Chisq")

vif(m6)
par( mfrow=c(2,2))
plot( m6, id.n=0 )
```

### 3.3 Include factor main effects and interactions to the best numeric model & Residual Analysis

```
#Null model
m0 <- glm(satisfaction ~ 1, family=binomial, data=train)
summary(m0)

#Numerical variables
m1 <- glm(satisfaction ~ Age + Flight.Distance + Arrival.Delay.in.Minutes + Departure.Delay.in.Minutes,
          data=train)
summary(m1)
vif( m1 )
par( mfrow=c(2,2))
plot( m1, id.n=0 )

# transformations
m2 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance + Arrival.Delay.in.Minutes + Departure.Delay.in.Minutes,
          data=train)
par( mfrow=c(2,2))
plot( m2, id.n=0 )

m3 <- glm(satisfaction ~ poly(Age,3) + Flight.Distance + Arrival.Delay.in.Minutes + Departure.Delay.in.Minutes,
          data=train)
par( mfrow=c(2,2))
plot( m3, id.n=0 )

anova(m2,m3) # m3 better than m2

anova(m1,m3) # m3 better than m1

m4 <- glm(satisfaction ~ poly(Age,3) + log(Flight.Distance) + log(Arrival.Delay.in.Minutes+1) + log(Departure.Delay.in.Minutes+1),
          data=train)

anova(m3, m4) # m4 worst than m3
AIC(m1,m2,m3, m4) # m3 small AIC -> m3 is the best at the moment

# All variables
summary(train)
m5 <- glm(satisfaction ~ poly(Age,3) + Customer.Type + Gender + Type.of.Travel + Class + Flight.Distance,
          data=train)
summary(m5)

anova(m3, m5, test="Chisq") # m5 better than m3
```

```

waldtest(m3, m5, test="Chisq")

# with new factors
m6 <- glm(satisfaction ~ poly(Age,3) + Customer.Type + Gender + Type.of.Travel + Class + Flight.Distance

anova(m5,m6) # m6 better than m5
AIC(m3,m5,m6)

m7 <- step(m6)
# satisfaction ~ poly(Age, 3) + Customer.Type + Type.of.Travel +
#   Class + Inflight.wifi.service + Departure.Arrival.time.convenient +
#   Ease.of.Online.booking + Gate.location + Online.boarding +
#   Seat.comfort + Inflight.entertainment + On.board.service +
#   Leg.room.service + Baggage.handling + Checkin.service + Inflight.service +
#   Cleanliness + Arrival.Delay.in.Minutes + Departure.Delay.in.Minutes +
#   AgeGroup + DelayedArrival

anova(m6,m7)
AIC(m6,m7) # m7 better but based on the anova results they are the same (NO SE SI ESTA BE AIXO)

# Add interactions with f.flightDist
m8 <- glm(satisfaction ~ poly(Age,3) + Customer.Type + Type.of.Travel +
  Class + Inflight.wifi.service + Departure.Arrival.time.convenient +
  Ease.of.Online.booking + Gate.location + Online.boarding +
  Seat.comfort*f.FlightDist + Inflight.entertainment + On.board.service +
  Leg.room.service*f.FlightDist + Baggage.handling + Checkin.service + Inflight.service +
  Cleanliness + Arrival.Delay.in.Minutes + Departure.Delay.in.Minutes+
  AgeGroup + DelayedArrival, family = binomial, data = train)
summary(m8)

anova(m7,m8)
AIC(m7,m8) # m8 better # 2135.30

vif(m8)
# si en canvi de Poly(Age, 3) poso Age el AIC augmenta a 2140 pero el valor del GVIF baixa de 43 a 5
# si borro el Departure.Delay.in.Minutes el AIC augmenta a 2138 pero el valor del GIF de arrival.Delay.

# Add transformation with DelayedArrival
m9 <- glm(satisfaction ~ poly(Age,3) + Customer.Type + Type.of.Travel*DelayedArrival +
  Class + Inflight.wifi.service + Departure.Arrival.time.convenient +
  Ease.of.Online.booking + Gate.location + Online.boarding +
  Seat.comfort*f.FlightDist + Inflight.entertainment + On.board.service +
  Leg.room.service*f.FlightDist + Baggage.handling + Checkin.service + Inflight.service +
  Cleanliness + Arrival.Delay.in.Minutes + Departure.Delay.in.Minutes+
  AgeGroup, family = binomial, data = train)
AIC(m8,m9) # m9 better -> AIC = 2128

# Add transformation with AgeGroup --> NO WORTH IT
m10 <- glm(satisfaction ~ poly(Age,3) + Customer.Type + Type.of.Travel*DelayedArrival +
  Class + Inflight.wifi.service + Departure.Arrival.time.convenient +
  Ease.of.Online.booking*AgeGroup + Gate.location + Online.boarding*AgeGroup +
  Seat.comfort*f.FlightDist + Inflight.entertainment*AgeGroup + On.board.service +
  Leg.room.service*f.FlightDist + Baggage.handling + Checkin.service + Inflight.service +

```

```

Cleanliness + Arrival.Delay.in.Minutes + Departure.Delay.in.Minutes, family = binomial, data = tra
AIC(m9,m10) # m9 better -> 2128
par(mfrow=c(2,2))
plot(m9)
plot(m4) # model with just numerical variables
summary(m9)

```

Now after some preliminary model exploration, step-wise regression will be performed in order to find the best possible model according to Akaike's criterion.

```
mAIC <- stepAIC(m7)
```

```
mAIC
```

### 3.2 Effects study

```
plot(allEffects(m7))
```

### 3.3 Residual analysis and Influential data study

```

# # Use for glms
# residualPlots(m7)
# influencePlot(m2, id.n=5)
#
# # 913, 951 can be considered influent data (However, this doesn't equal to being outlier or noise...
# Boxplot(cooks.distance(m7), id=list(labels=rownames(dfwork)))

```

### 3.4 Final model interpretation

The final model that obtained after a long model discovery is the following, which has a pseudo-coefficient of determination (McFadden) of X.

TODO: Explicar amb una formula de latex la interpretacio del model final amb logit i probabilitats i tal

```

#
# best_model <- m1
#
# #Summary of the parameters
# summary(best_model)
# #Pseudo coefficient of determination R^2
# psuedoR2 <- pR2(best_model)["McFadden"]
# #Residual analysis and influent data
# residualPlots(best_model)
#
# #Evaluation of the model using train data (Confusion matrix and AUC)
#
# # Get predicted probabilities and classifications
# predicted_prob <- predict(best_model, newdata = train, type = "response")

```

```

# #Classification threshold
# threshold <- 0.5
# predicted_class <- ifelse(predicted_prob > threshold, 1, 0)
#
# # Confusion Matrix
# actual_class <- train$satisfaction
# conf_matrix <- confusionMatrix(as.factor(predicted_class), as.factor(actual_class));conf_matrix
#
# # Extract evaluation metrics
# accuracy <- conf_matrix$overall["Accuracy"]
# precision <- conf_matrix$byClass["Precision"]
# recall <- conf_matrix$byClass["Recall"]
# f1_score <- 2 * (precision * recall) / (precision + recall)
#
# cat("Accuracy:", accuracy, "\n")
# cat("Precision:", precision, "\n")
# cat("Recall:", recall, "\n")
# cat("F1 Score:", f1_score, "\n")
#
# # AUC and ROC Curve
# roc_curve <- roc(actual_class, predicted_prob)
# auc_value <- auc(roc_curve)
# cat("AUC:", auc_value, "\n")
#
# # Plot the ROC Curve
# plot(roc_curve, col = "blue", main = "ROC Curve", lwd = 2)
# abline(a = 0, b = 1, lty = 2, col = "red") # Add diagonal reference line

```

### 3.5 Model evaluation

Evaluation of the aforementioned model with unseen data will be performed using various categorical data criteria such as Precision, Recall, Accuracy, F1-score and the AUC/ROC curve.

## 4 Conclusion and Further Work

TODO: Falten totes les conclusions :)

**Extensissimo prolongado relato**